

Application Note: Flex Sensor Subsystem

Joanna Chronis
Student Number: 2817173
University of the Witwatersrand
2817173@students.wits.ac.za

Abstract

A subsystem for a crypto password-protected box was required to power a green LED and a relay when a correct bend angle was acquired on a SEN-10264 flex sensor. A gain of 4.5 was applied to the output of a Wheatstone Bridge containing the sensor, and a comparator ladder containing 10 LEDs was powered using this output to show different bend angles as the flex sensor was bent. A green LED switches on after a 2-second Delay if the bend angle was within the required angle bracket of 80° , and a red LED switches on if the angle was not within the bracket. The subsystem was simulated using LTSpice and physically implemented on a breadboard. The measured results were a valid comparison against the simulated results and thus prove the validity of the design, confirming the operation of the subsystem.

Index Terms

Comparator Ladder; Operational Amplifier; SEN-10264 flex sensor; LTSpice simulation; DC Operating point ; Gain

I. INTRODUCTION

This application note presents a subsystem of a larger crypto-password-protected box system. This will further be referred to as the subsystem. The subsystem is required to gather an input through bends of a flex sensor at different angles, with a specified angle at 80° to be the required angle for a "correct" result. The subsystem needs to have password verification logic to output a valid response to whether the correct angle has been met or not at the end of a 2-second delay, which will be implemented using a simple RC circuit. A SEN-10264 Flex sensor [1] within a Wheatstone Bridge is used to produce an input from the different bending angles, which in turn sends a voltage through a differential amplifier, powering a series of 10 LEDs within a comparator ladder to display the angle the sensor has been bent to. The result of the comparator ladder is then input into a logic system consisting of an AND gate. This toggles either the red or green LED display after the 2-second delay in order to depict whether the correct angle was inputted or not. If the correct angle is met, a relay will power on and thus provide power to the next subsystem in the overall system. A block diagram of the flex subsystem is shown in Figure 1. The three objectives that need to be met over this subsystem is the design, simulation and implementation of it. Section 2 addresses objective (i) and discusses the schematic circuit and component design of the flex sensor subsystem. Section 3 addresses objective (ii) by undergoing an LTSpice DC Operating point simulation in order to observe the validity and performance of the abovementioned subsystem. Section 4 addresses objective (iii) by presenting physical oscilloscope measurements taken from a built version of the subsystem, in order to ensure accuracy of the simulated design and confirm the subsystem works as simulated. Section 5 provides a conclusion to the application note.

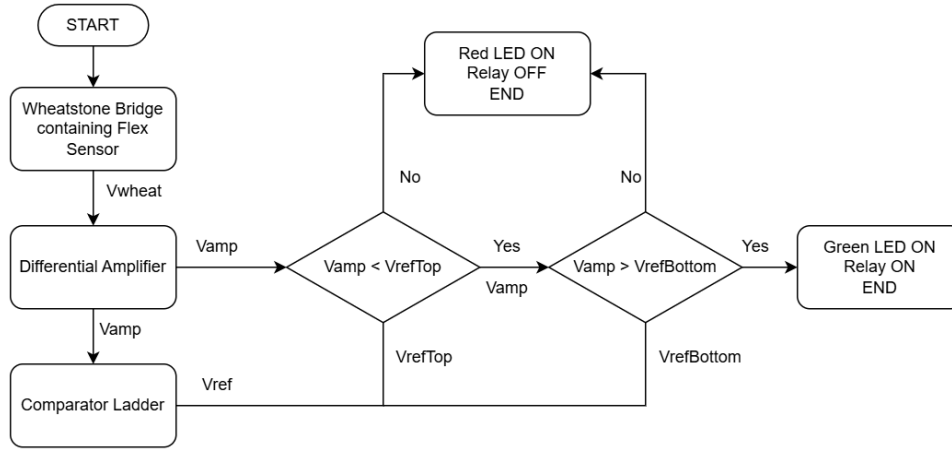


Fig. 1: Subsystem block diagram with correct angle validation logic.

II. SYSTEM DESIGN

The Subsystem schematic design is shown in Figure 2 and consists of a power supply of 5 V; a non-inverting differential amplifier (A_1); an example comparator from the subsystem comparator ladder; and an RC delay that charges when the correct flex angle is met. The entire subsystem is powered by the 5 V power supply and does not require negative voltages; thus, any negative supply rails are tied to ground. The power supply provides voltage to the Wheatstone bridge containing the flex sensor, which in turn provides V_1 and V_2 . V_1 and V_2 are the negative and positive output nodes of the Wheatstone bridge, respectively. The voltage difference of these nodes will further be referred to as $V_{wheatstone}$. Bending the flex sensor to certain angles will alter the output of the differential amplifier as the input of the amplifier is increasing or decreasing, depending on the angle of the flex sensor, as depicted in Figure 3. Due to the small difference in resistance between each angle bracket, the voltage provided by the amplifier needs to be much larger than that of the Wheatstone bridge, as $V_{wheatstone}$ does not

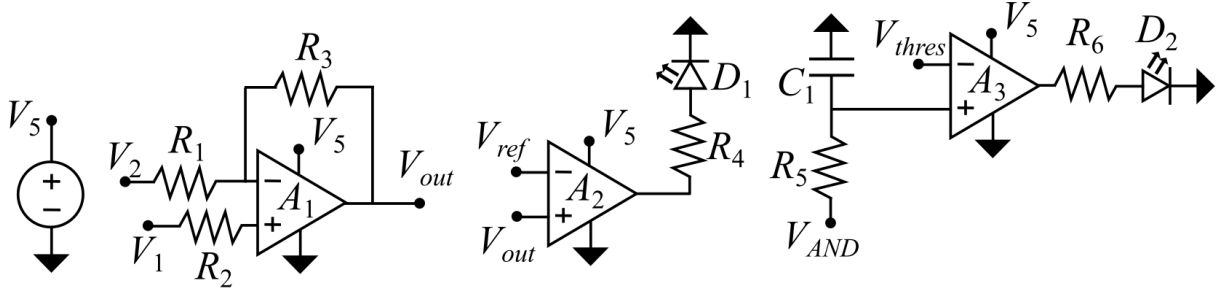


Fig. 2: System schematic showing the power supply, non-inverting differential amplifier, comparator ladder operational amplifier and correct angle RC delay.

provide nearly enough voltage to be able to accurately decipher between neighbouring angle brackets. Due to the need of a larger voltage, a gain of $G = 4.5$, expressed by

$$G = 1 + \frac{R_3}{R_1}, \quad (1)$$

is applied to $V_{wheatstone}$ via (A_1), by setting $R_1 = 10 \text{ k}\Omega$ and $R_3 = 35 \text{ k}\Omega$. This amplified Voltage, denoted V_{out} in the schematic, is then used in the comparator ladder and RC delay circuit. The comparator ladder consists of 10 comparators, each with a unique V_{ref} listed in Table I, which in turn switches on a series of LEDs as the flex sensor is bent and V_{out} surpasses V_{ref} .

TABLE I: Reference voltages for 10 LED comparator Ladder

Angle Bracket ($^\circ$)	Vref (V)
0	0.00
10	0.34
20	0.84
30	1.68
40	1.92
50	2.88
60	3.52
70	4.28
80	4.69
90	4.89

Once the flex sensor is bent into the correct angle bracket, the RC delay is powered via an AND gate with logic in Table II. It is clear to see in this table that the RC delay will only begin to charge if the bend angle is between 80° and 90° . In order for the delay to be roughly two seconds long, the values of R_5 and C_1 were determined using

$$\tau = RC, \quad (2)$$

giving the values of $R_5 = 100 \text{ k}\Omega$ and $C_1 = 220 \mu\text{F}$. Once the capacitor has charged to a voltage $2/3$ times the size of 5 V, denoted V_{Thres} and determined via a simple two-resistor voltage divider, LED D_2 powers and indicates the correct angle has been input.

TABLE II: Logic Table for Password Verification

Angle Bracket ($^\circ$)	Vref (V)	AND Gate Output (Password Verified)
0	0.00	FALSE
10	0.34	FALSE
20	0.84	FALSE
30	1.68	FALSE
40	1.92	FALSE
50	2.88	FALSE
60	3.52	FALSE
70	4.28	FALSE
80	4.69	TRUE
90	4.89	FALSE

III. SIMULATION

A DC operating point simulation was performed using LTSpice to confirm the design specifications set out in the design section.

Looking at the sensor characteristics in Figure 3, it is clear to see that the flex sensor could be simulated using a variable resistor in place of the sensor, as bending the flex sensor results in an increase in resistance. Therefore, the DC operating point simulation was run by varying the resistance of a variable resistor in increments that aligned with those of the resistances graphed in Figure 3. This simulation of the sensor allowed the value of gain needed by the amplifier to be determined, without wearing out the physical flex sensor provided for implementation. By looking at the simulation results in Figure 4, the simulated amplifier output, visualised by trace V_{out} , is a clear depiction

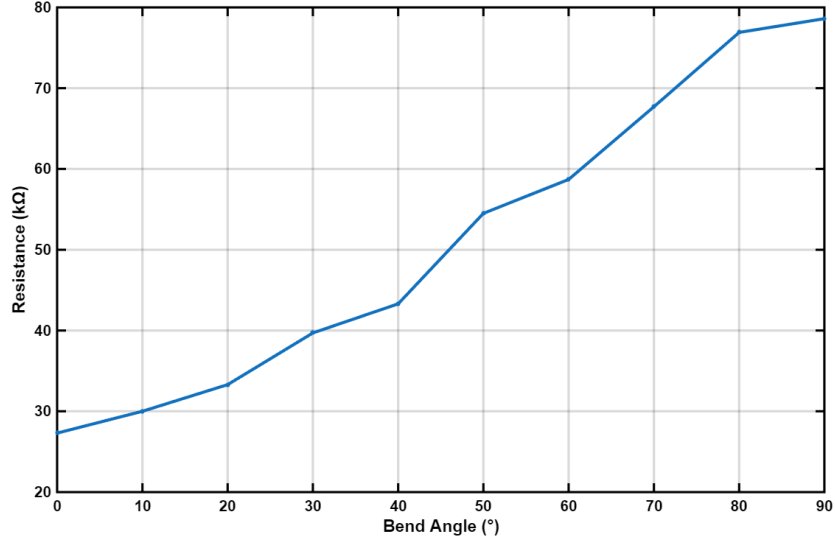


Fig. 3: SEN-10264 Flex Sensor Characteristics.

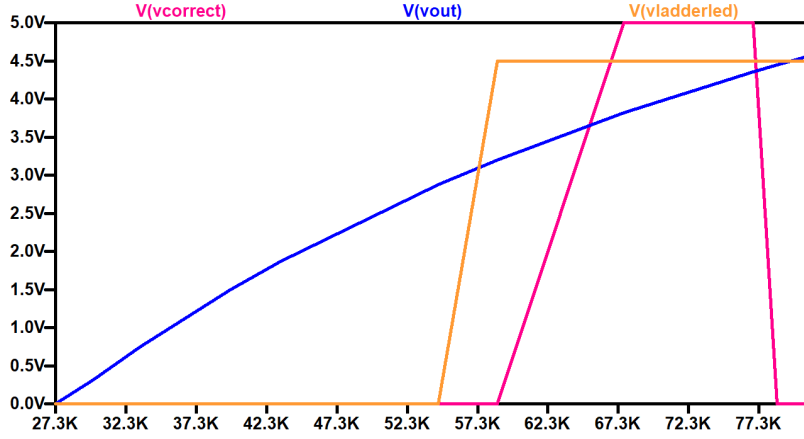


Fig. 4: LTSpice Simulation results.

of the calculated gain as it provides a voltage high enough to be used by the rest of the subsystem. This amplified voltage range can be used within the comparator ladder with greater accuracy than the voltage range that was not amplified.

Trace $V_{\text{LadderLED}}$ shows that the LEDs within the comparator ladder are triggered and remain on when their reference voltages are reached and surpassed, which verifies the design imaged in Figure 1. Although this trace only shows a single LED functioning, the other remaining 9 LEDs all work on the same logic. Trace V_{Correct} simulates the entry and exit of the flex output in the angle bend window needed in order for the correct angle to be determined, depicted by the "True" case in Table II. When the angle is bent enough to be above the lower limit voltage but below the higher limit, V_{Correct} is High (4.5 V) and is Low (0 V) when this criterion is not met.

IV. IMPLEMENTATION

The circuit in Figure 1 was implemented and built across three breadboards, with a power supply of 5 V powering all positive rails. The measurements in Figure 5 were taken using 1x oscilloscope probes. All traces were exported to CSV files for external plotting using Python. Due to a shortage of comparators, which were more ideal for this subsystem than Operational Amplifiers (Op Amps), Lm324 Op Amps [2] were used due to their quad space efficiency on the breadboards, availability, single power supply requirements, and wide voltage range. The fact that the Lm324 cannot swing to the positive rail will not impact implementation as it was already accounted for in the design of the subsystem through a high enough Gain. The use of 10 kΩ trimmer potentiometers (trim pots) ensured that the correct V_{ref} s needed for the comparator ladder could be obtained, as well as the thresholds for the password verification logic. Trim pots were chosen due to their easy adjustability, as set resistor values would be tedious, as flex sensors do not have exact set angle resistances, but rather fluctuate slightly depending on how much they have been bent in the past. Figure 5 shows the measured oscilloscope traces of the physical circuit. The measured trace V_{out} of the amplifier acts within a range of roughly 0.1V to 3.5V. This discrepancy in voltage compared to the simulated voltage comes from the fact that Lm324s do not swing to the positive rail as mentioned above, but rather swing to roughly 1.5 V below the positive rail. Despite this difference, the output of the comparator ladder, $V_{\text{LadderLED}}$, and the output of the input verification, V_{Correct} , match the behaviour of the simulated circuit. Although actual voltages may vary due to the discrepancy mentioned earlier, the overall design is validated through a visible correlation between simulation and implementation results. The amplifier achieves the required gain needed for $V_{\text{LadderLED}}$ and V_{Correct} to function as needed, even though the real case V_{out} was not as high as the voltage simulated.

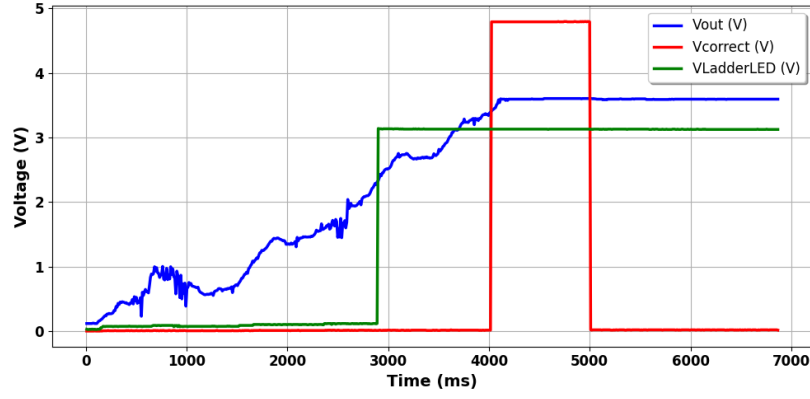


Fig. 5: Implementation results recorded on an oscilloscope and plotted from a CSV file.

V. CONCLUSION

This application note presented a subsystem of a crypto password-protected box that powers a relay and green LED when a flex sensor is bent to a specific angle, namely 80° . The subsystem design consists of a Wheatstone bridge containing the flex sensor; a differential amplifier with $G = 4.5$ needed to increase the voltage difference produced by the Wheatstone bridge into a range of 0 V – 3.5 V; a comparator ladder that powers different LEDs depending on voltage; and a 2-second RC delay used to confirm the correct angle input. LTSpice simulations confirmed correct voltage amplification gain via the differential amplifier and correct behaviour of the various LEDs need to show the bend angle and correct status. The physical implementation on a breadboard reproduced the simulated results, although with a 1 V discrepancy between the simulated amplifier and physical amplifier, due to the use of LM324 operational amplifiers. The physical implementation functioned as needed in order to input and verify a specified angle of 80° as correct, thus showing the design meets the subsystem specifications.

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